BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 149 · Lesson 149 · Client–Server and Peer-to-Peer Network Models

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which device connects several wireless users to an existing school LAN?

Q2. Which storage type is volatile?

Q3. Which layer in TCP/IP handles IP routing?

Q4. What data structure follows FIFO?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

client–server — Managed service provided by server to requesting clients.

peer-to-peer — Peers directly share services/resources.

centralisation — Services controlled from a smaller set of authorities.

scalability — How well a system handles growth in load/users.

New learning

Client–server — Many clients request services from a managed central server.

Peer-to-peer — Each participating peer may offer and consume resources.

Select architecture — Balance availability, management, trust and workload distribution.

Worked example

A school stores assessment marks requiring access control, backups and a shared authoritative copy. Recommend a model.

1. Teachers send requests — Identify the facts and make an initial prediction.

2. Central marks database — Explain the mechanism or trace the intermediate stage.

3. Access control + backup — Client–server suits managed central data, controlled permissions, auditing and backups. A server failure can affect many clients, so redundancy and resilient backups matter. Peer-to-peer file sharing is harder to manage consistently.

We Do · Practise with guidance

A small group distributes a large open-licensed software image to many volunteers. Is peer-to-peer appropriate?

Peers hold file blocks

Identify the facts and make an initial prediction.

Peers share download load

Explain the mechanism or trace the intermediate stage.

Verify authentic file

A P2P distribution model can share upload load among peers, reducing demand on one central server. Clients must verify file authenticity with trusted signatures/hashes; peers may be untrusted or offline.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Define client, server and peer in the context of a network service. [6]

Question 2

Explain two benefits and one limitation of client–server for a school's student database. [6]

Question 3

Compare P2P and client–server for distribution of a large video archive, including integrity checks. [8]

Exit Ticket · Answer before checking

A network using client–server can never contain any peer-to-peer activity.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A wireless access point.

Q2. RAM.

Q3. The internet layer.

Q4. A queue.

You Do pages · Independent questions

Q1. Client requests services [2]; server provides managed services [2]; peer can act as client/provider [2].

Q2. Centralised data/controls [2], easier backups/updates [2], single service outage/cost [2].

Q3. P2P resource sharing [2], client-server central control [2], availability/scalability [2], trustworthy verification/justified recommendation [2].

Exit Ticket · Model answer

False. Hybrid architectures can use multiple patterns.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: